

1 Conclusions

Chapter 1 framed this thesis around an auditable six-stage engineering pipeline (from a raw optical video archive to a usable design-triage tool) to address the challenges of marine multi-fuel pilot injection development. This closing chapter summarizes the core novelties introduced by this work, synthesizes the primary scientific and engineering findings, outlines the operational boundaries of the current prototype, and proposes concrete directions for future research.

1.1 Key Novelties and Methodological Contributions

This thesis introduces four principal novelties bridging optical diagnostics and surrogate modeling:

1. **Censoring-Aware Target Parameterization (SGQR):** A reduced-order Sigmoid-Gated Quarter-Root fitting formulation that compresses noisy plume trajectories into smooth, differentiable targets while explicitly accommodating right-censored observations caused by a finite camera field of view. This prevents data-driven models from mistaking optical image window exit for physical spray saturation.
2. **Curriculum-Guided Staged Uncertainty Learning with Knowledge Distillation:** A three-stage training strategy combining mean fitting, conditional dispersion learning, and regime-aware teacher-student knowledge distillation. This framework allows the neural surrogate to output both penetration predictions and heteroscedastic uncertainty bounds under right-censored observation limits.
3. **Probabilistic Spray–Wall Impingement Screening Framework:** A rapid geometric interface that maps predicted penetration distributions onto continuous wall-contact probabilities ($P_{\text{hit}}(t)$) and cumulative exposure scores (E). This shifts design triage from binary deterministic pass/fail metrics to risk-quantified screening prior to high-cost computational fluid dynamics (CFD) or engine testing.
4. **Automated Video-to-Observable Processing Chain:** A high-throughput, CUDA-accelerated processing pipeline capable of turning raw multi-hole Mie scattering video archives into plume-wise temporal observables with full end-to-end auditability and reproducible checkpointing.

1.2 Main Findings and Engineering Insights

The experimental evaluations and benchmarking detailed in Chapters 4 and 5 yield the following key findings:

- **Substantial Superiority over Empirical Physical Baselines:** On the primary uncensored CDF evaluation protocol (540,367 points across

9 nozzle campaigns), the three-stage multilayer perceptron (MLP) surrogate achieves a 5-seed mean root mean squared error (RMSE) of 4.19 mm and mean absolute error (MAE) of 2.98 mm. This represents a $> 54\%$ and $> 59\%$ reduction in point-wise RMSE compared to refitted and calibrated Hiroyasu–Arai (10.19 mm) and Naber–Siebers (9.21 mm) physical baselines, demonstrating that data-driven surrogates capture transient pilot injection development far better than classical empirical spray laws.

- **Architectural Trade-off Between Point Accuracy and Coverage Safety (MLP vs. SVGP):** A comparison between the three-stage MLP backbone and a Sparse Variational Gaussian Process (SVGP) benchmark reveals a fundamental accuracy–coverage trade-off. While the SVGP attains slightly better point-wise accuracy (4.09 mm RMSE), the MLP provides markedly more conservative $1\sigma/2\sigma$ uncertainty coverage (88.8%/99.0% vs. 71.7%/94.6%). For engineering risk screening, the MLP backbone is therefore safer and more reliable, as under-estimating uncertainty could lead to false-negative wall impingement assessments.
- **Out-of-Distribution Transferability and Inductive Bias Limits:** Leave-one-nozzle-out (LONO) cross-validation demonstrates strong transferability across geometrically similar multi-hole nozzle designs (RMSE ≈ 5.95 mm–7.00 mm when excluding the out-of-family Nozzle 0). However, performance degrades on Nozzle 0 (RMSE ~ 27 mm–35 mm), identifying the boundaries of inductive bias when extrapolating to entirely novel injector hardware families.

Table ?? summarizes the final validated surrogate performance across both in-domain and out-of-distribution evaluation protocols.

Evaluation protocol	Three-stage MLP		SVGP	
	RMSE [mm]	$1\sigma/2\sigma$	RMSE [mm]	$1\sigma/2\sigma$
In-domain, uncensored CDF	4.185	0.888/0.990	4.087	0.717/0.946
LONO, 5-fold mean	12.55	0.575/0.770	10.16	0.628/0.821
LONO, 4-fold (excl. Nozzle0)	7.00	0.696/0.923	5.95	0.697/0.890

Table 1: Final validated surrogate scorecard comparing the production three-stage MLP against the SVGP baseline on the primary uncensored CDF protocol (540,367 points; 5-seed means) and leave-one-nozzle-out (LONO) transfer (5-fold means). The MLP provides consistently conservative coverage suited for screening; the SVGP offers tighter point accuracy.

1.3 Current Prototype Boundaries

While the workflow successfully achieves its core objectives, two primary operational boundaries define the current prototype:

- **Cold Non-Reacting Envelope Scope:** The surrogate models the liquid-phase Mie-defined spray envelope under room-temperature, non-reacting constant-volume vessel conditions. It provides an early conservative proximity bound rather than a phase-resolved predictor for hot evaporative liquid impingement or combustion dynamics.
- **Geometric Wall Interface Assumption:** The probabilistic wall-hit metric $P_{\text{hit}}(t)$ currently relies on a simplified 3D geometric projection and static piston boundary, which serves as a workflow demonstration and requires direct empirical or CFD validation under real engine geometry.

1.4 Next Development Priorities

Future enhancements to extend this workflow should focus on four key priorities:

1. **Hot CFD and Experimental Validation:** Validate predicted wall-contact probabilities $P_{\text{hit}}(t)$ and exposure scores E against matching high-temperature evaporative CFD simulations and optical spray-wall interaction experiments.
2. **Learnable 2D Spray Envelope Model:** Replace the current constant-cone-angle assumption with a learnable two-dimensional spatiotemporal envelope model to capture dynamic plume spreading.
3. **Condition-Clustered Uncertainty Bounds:** Incorporate condition-level and recording-level hierarchical clustering into confidence interval calculations to account for temporal autocorrelation among frame-wise samples.
4. **Transient Nozzle-Conditioned Priors:** Introduce nozzle-conditioned transient priors into the surrogate backbone to improve predictions for complex step-wise penetration behaviors (such as Nozzle 3) and novel hardware families (Nozzle 0).

1.5 Closing Remarks

By carrying an auditable, six-stage engineering discipline end-to-end—from raw optical video archives to uncertainty-aware trajectory surrogates and risk screening—this work establishes a practical methodology for re-using existing experimental data. The resulting workflow provides an efficient, low-cost screening layer that narrows the candidate space for marine multi-fuel pilot injectors before initiating resource-intensive CFD simulations and engine validation.